

Beacon Frame Analysis for Rogue AP Detection

Akintoye Oyedola

School of Computing, University of North Florida

CNT 6519 Wireless Network Security



Motivation & Problem Statement

Rogue access points (APs), including evil-twin devices that mimic legitimate Wi-Fi networks, allow attackers to intercept traffic or steal user credentials in enterprise WLAN environments. Detecting these attacks is difficult because malicious APs can replicate legitimate identifiers such as SSIDs and BSSIDs [3, 2]. Existing detection approaches often rely on **active probing, client monitoring, or infrastructure-based** intrusion prevention systems. These methods introduce additional overhead costs or privacy concerns [1]. I investigate whether passive beacon-frame and probe-response analysis alone, using structural and signal features from 802.11 management frames, can reliably detect rogue APs with low false-positive rates in realistic wireless environments [6].

System Design

A passive rogue AP detection pipeline analyzes IEEE 802.11 management frames captured from a wireless monitoring interface, enabling continuous background monitoring without active probing:

- Beacon Capture Layer:** A monitor-mode Wi-Fi adapter (Qualcomm Atheros AR9271) captures raw 802.11 management frames across 8 labeled sessions using airodump-ng, generating PCAP traces totaling 6,607 packets across sessions S1–S8.
- Feature Extraction Layer:** Python 3 scripts using **Scapy** parse raw frame control bytes to extract structural and signal features including SSID/BSSID consistency, sequence number deltas, inter-frame timing, and RSSI statistics [2].
- Detection Engine:** Rule-based anomaly detection (SSID/BSSID mismatch, RSSI range threshold, hidden SSID flag) and supervised machine learning classifiers implemented in **Scikit-learn** [3].
- Analysis and Output Layer:** Classification metrics including detection rate, false alarm rate, precision, recall, and ROC curves assess system performance [6].

The experimental pipeline, from hardware capture to machine learning evaluation, is illustrated in Fig 1

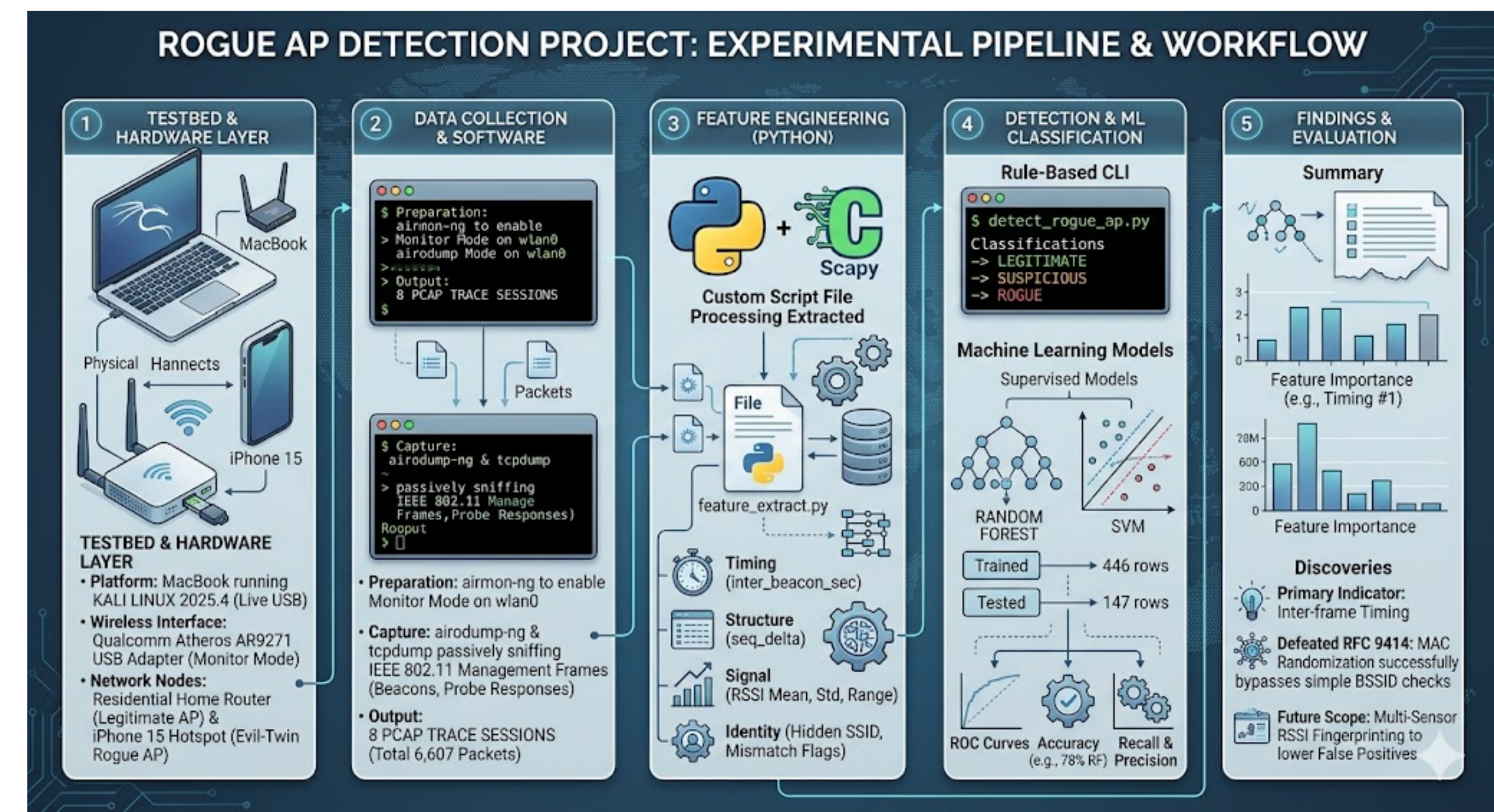


Fig. 1. This visualization illustrates the workflow from hardware capture (Kali Linux/Atheros AR9271) to machine learning classification (Random Forest/SVM). Image generated by Gemini 3 Flash based on author's experimental data [4]. Graphic generated by Google Gemini, [March 25, 2026], <https://gemini.google.com/>.

Key Technologies & Components

Wireless Packet Capture

- **airodump-ng:** Captures IEEE 802.11 management frames in monitor mode across wireless channels.
- **tcpdump:** Records raw PCAP traces for offline analysis.

Packet Analysis & Feature Extraction

- **Scapy:** Parses management frames via raw frame control byte inspection; extracts BSSID, SSID, sequence number, RSSI, and inter-frame timing.
- **Wireshark:** Validates captured packets and verifies frame structure.

Data Processing

- **Python 3.13:** Primary language for packet parsing and feature automation.
- **Pandas:** Organizes extracted features into structured labeled datasets.

Detection Models

- **Scikit-learn:** Implements Random Forest and SVM classifiers with `class_weight='balanced'`.
- **Rule-Based Detection:** Flags SSID/BSSID mismatches, RSSI anomalies, and hidden SSIDs.

Evaluation Metrics

- Accuracy, Precision, Recall, F1-score
- False Positive Rate (FPR)
- ROC Curve and AUC
- Confusion Matrix

Data Collection & Implementation

Session Condition		Frames	Label	Role
S1	Legitimate-only baseline	207	0	Training
S2	Evil-twin, different channel	62	1	Training
S4	Legitimate-only, different time	192	0	Training
S5	Evil-twin, distance variation	287	1	Training
S6	SSID-cloaked legitimate AP	182	−1	Qualitative
S7	Legitimate-only repeat	218	0	Test
S8	Evil-twin repeat	121	1	Test

Experimental session matrix: capture conditions, frame counts, labels, and roles for evil-twin AP detection.

wlan.fc.type == 0x0 && wlan.fc.type_subtype == 0x8 && wlan.fixed.capabilities.privacy == 0					
No.	Time	Source	Destination	Protocol	Length Info
16...	152437.97...	Tp-LinkT_16:87:...	Broadcast	802...	120 Beacon frame, SN=941, FN=0, Flags=...
16...	152438.07...	Tp-LinkT_16:87:...	Broadcast	802...	120 Beacon frame, SN=942, FN=0, Flags=...
16...	152438.17...	Tp-LinkT_16:87:...	Broadcast	802...	120 Beacon frame, SN=943, FN=0, Flags=...
16...	152438.28...	Tp-LinkT_16:87:...	Broadcast	802...	120 Beacon frame, SN=944, FN=0, Flags=...
17...	152438.38...	Tp-LinkT_16:87:...	Broadcast	802...	120 Beacon frame, SN=945, FN=0, Flags=...
17...	152438.48...	Tp-LinkT_16:87:...	Broadcast	802...	120 Beacon frame, SN=946, FN=0, Flags=...
17...	152438.58...	Tp-LinkT_16:87:...	Broadcast	802...	120 Beacon frame, SN=947, FN=0, Flags=...
18...	152438.69...	Tp-LinkT_16:87:...	Broadcast	802...	120 Beacon frame, SN=948, FN=0, Flags=...
18...	152438.79...	Tp-LinkT_16:87:...	Broadcast	802...	120 Beacon frame, SN=949, FN=0, Flags=...
18...	152438.89...	Tp-LinkT_16:87:...	Broadcast	802...	120 Beacon frame, SN=950, FN=0, Flags=...
Radiotap Header v0, Length 36					
802.11 radio information					
IEEE 802.11 Beacon frame, Flags:C					
IEEE 802.11 wireless LAN					
Fixed parameters (12 bytes)					
Tagged parameters (44 bytes)					
Tag: SSID parameter set: SecurityTube_Open					
Tag Number: SSID parameter set (0)					
Tag length: 17					
SSID: SecurityTube_Open					
Tag: Supported Rates 1(B), 2(B), 5.5, 11, [Mbit/sec]					

Fig. 2. Sample IEEE 802.11 beacon frame from Wireshark with management frame fields. [5]

Experiment Setup

- 8 labeled capture sessions conducted across 2 sittings on different days using a Kali Linux live environment with a monitor-mode USB Wi-Fi adapter.
- Sessions ran 10 minutes each; raw PCAP files totaled 1.4–2.1 MB per session across 6,607 total packets.
- Dataset included frames from a legitimate home router (BSSID 54:4e:67:34:56:4d) and an iPhone 15 Personal Hotspot configured as an evil-twin AP with identical SSID (ATTNg4VMGJ).
- 593 labeled management frames extracted after filtering to known BSSIDs: 123 legitimate (label 0), 470 rogue (label 1), 182 cloaked (label −1, qualitative only).
- Extracted features fed into Scikit-learn Random Forest and SVM classifiers with session-stratified train/test split.

Key Findings

- **593 labeled frames** extracted across 7 sessions from 2 capture sittings on different days.
- **SSID cloaking** (S6) eliminated SSID-based detection entirely, forcing reliance on RSSI and timing features alone [3].
- **False positive rate** on legitimate frames exceeded the 5% target from Zhang et al. [6], due to class imbalance and single-device legitimate AP dataset.

Performance Results

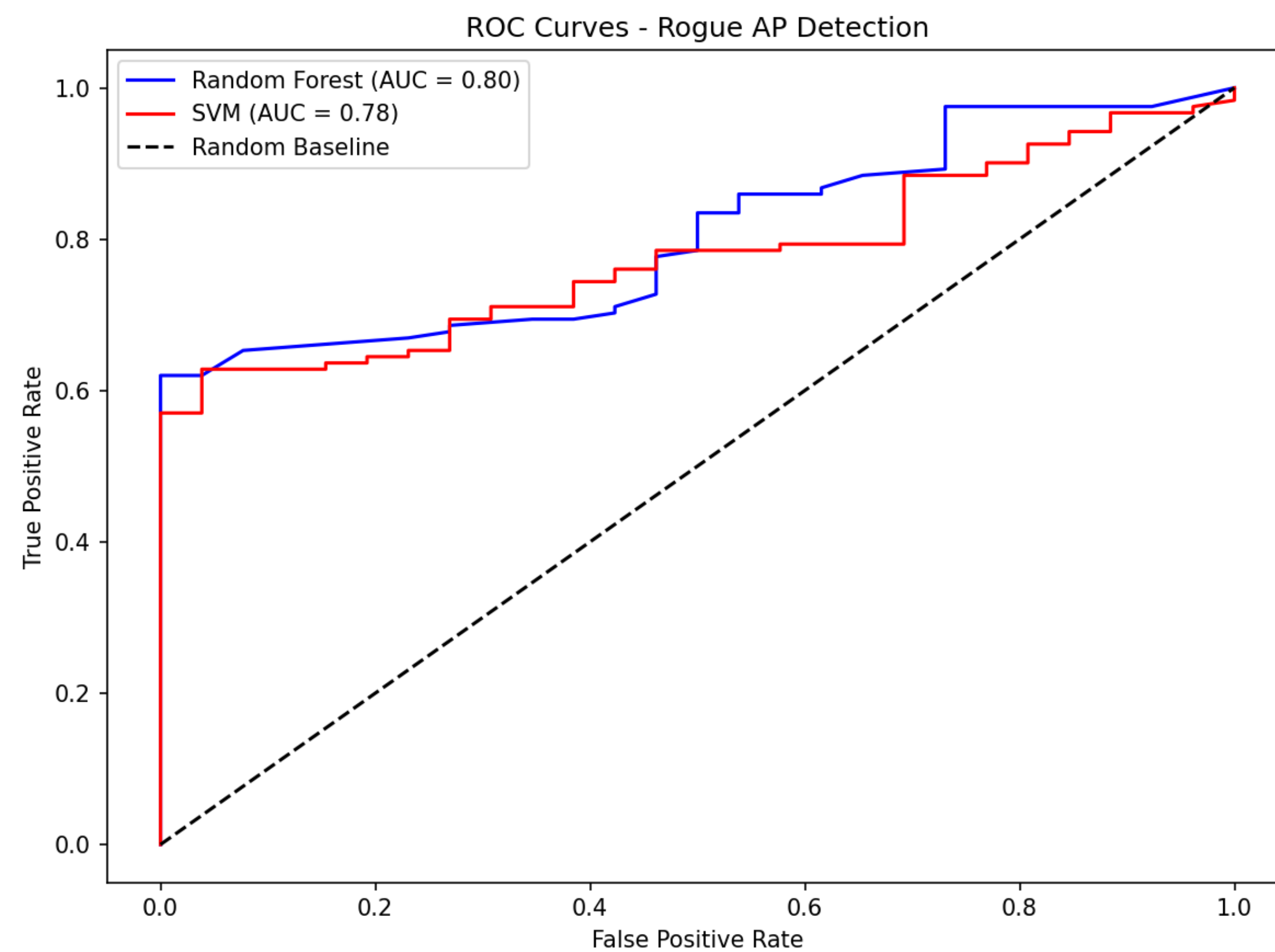
Classifier Comparison (Held-Out Test Set: S7, S8)

Classifier	Class	Precision	Recall	F1
Random Forest	Legitimate	0.39	0.50	0.44
	Rogue	0.89	0.83	0.86
	Accuracy	0.78		
SVM	Legitimate	0.26	1.00	0.42
	Rogue	1.00	0.48	0.57
	Accuracy	0.50		

Performance Results

Random Forest Feature Importance

Feature	Importance
inter_beacon_sec	0.8316
seq_num	0.1649
ssid_bssid_mismatch	0.0001
rss_i_mean, rss_i_std, rss_i_range	≈0.0000



ROC curves for Random Forest and SVM on held-out test set (S7, S8). Random Forest achieves higher AUC, reflecting stronger discrimination between legitimate and rogue AP frames.

Conclusions

- The **Random Forest classifier** achieved **83% recall** on the rogue class and **78% overall accuracy**, outperforming the SVM on all rogue-detection metrics.
- **Inter-frame timing** (importance: 0.83) was the dominant discriminating feature, reflecting structural differences between periodic router beacons and reactive iPhone probe responses.
- **MAC address randomization** (RFC 9414 [7]) defeated the SSID/BSSID mismatch rule: the iPhone hotspot's randomized BSSID prevented evil-twin flagging in both the rule-based script and the ML feature vector.
- **SSID cloaking** (S6) eliminated SSID-based detection entirely, consistent with Liu et al. [3], forcing reliance on RSSI and timing features alone.
- **False positive rate** on the legitimate class (≈50%) exceeded the 5% target from Zhang et al. [6], driven by class imbalance and single-device legitimate AP diversity.
- Future work should incorporate **multi-sensor RSSI fingerprinting**, beacon-focused (not probe-response-dominated) captures, and larger datasets with diverse rogue AP hardware to improve generalization.

References

- [1] B. Da, Z. Zhang, and X. Wang. “Rogue access point detection: A survey”. In: *IEEE Communications Surveys & Tutorials* 23.1 (2021), pp. 245–268. DOI: 10.1109/COMST.2020.3040256. URL: <https://doi.org/10.1109/COMST.2020.3040256>.
- [2] J. Lin, J. Wang, and T. Xu. “Detecting rogue access points via beacon frame analysis and machine learning”. In: *IEEE Transactions on Network and Service Management* 19.3 (Sept. 2022), pp. 2105–2118. DOI: 10.1109/TNSM.2022.3164421. URL: <https://doi.org/10.1109/TNSM.2022.3164421>.
- [3] H. Liu, S. Chen, and M. Zhou. “Position-based rogue AP detection using spatial signal properties”. In: *Journal of Network and Computer Applications* 221 (Jan. 2024), p. 103784. DOI: 10.1016/j.jnca.2023.103784. URL: <https://doi.org/10.1016/j.jnca.2023.103784>.
- [4] Akintoye Oyedola. *Rogue AP Detection Project: Experimental Pipeline & Workflow*. [Infographic]. Generated by Gemini 3 Flash based on data from *Wireless Network Security: Beacon Frame Analysis for Passive Rogue Access Point Detection*. University of North Florida, School of Computing. Mar. 2026.
- [5] Andrew Walding. *Capturing wi-fi beacon frames with WinFi*. Sept. 2025. URL: https://www.cellstream.com/2025/06/30/capturing-wi-fi-beacon-frames-with-wifi/?utm_source=chatgpt.com.
- [6] L. Zhang, Y. Yang, and C. Wang. “PRAPD: Passive rogue access point detection using RSSI fingerprinting”. In: *Proc. IEEE Global Communications Conference (GLOBECOM)*. 2018, pp. 1–7. DOI: 10.1109/GLOCOM.2018.8647145. URL: <https://doi.org/10.1109/GLOCOM.2018.8647145>.
- [7] J. C. Zúñiga, C. J. Bernardos, and A. Andersdotter. *Privacy considerations for IEEE 802.11 operating procedures*. Request for Comments 9414. IETF, Jan. 2023. URL: <https://www.rfc-editor.org/rfc/rfc9414.html>.